Reproducing Easy Data Augmentation: Does It Hold Up on New Text Classification Data?

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Abstract

Easy Data Augmentation (EDA) (Wei and Zou, 2019) makes extra training sentences with four random word-level edits and is reported to improve text classifiers, most of all on small training sets. I tried to reproduce that claim. On SST-2, using the authors' code and the CNN from the paper, I found no gain: EDA scored 0.5 to 0.7 points *below* normal training at 500, 2,000 and 5,000 sentences and level at the full set, and no difference was statistically significant. I then applied the method to two new datasets: the Emotion tweets, and 199 labelled Google Play reviews of the Woolworths and Coles apps that I collected. The results were mixed. EDA helped on Emotion at 500 training tweets (+2.1 accuracy) and at 5,000 (+2.0 macro-F1), did nothing at 2,000, and made no significant difference on the reviews. An LLM-as-judge check of 80 edited texts found that about 95% of edits kept their label, but fluency was only moderate (2.8 out of 5), with random swap the worst. Code, results and the review dataset (five reviews withheld) are public.

1 Introduction

Labelled text is expensive to produce, and classifiers trained on small datasets overfit. *Data augmentation* tackles this by creating altered copies of training examples that keep their labels. Computer vision uses flips and crops as a matter of course. In text, even a small edit can change what a sentence means, so augmentation is harder. Wei and Zou (2019) proposed Easy Data Augmentation (EDA): four random word-level edits that need no extra model and no extra data. They reported gains on five classification benchmarks, especially with few training examples.

This report describes my attempt to reproduce that result and to test whether it generalises. I make three contributions: (1) a replication of the paper's main experiment on SST-2 with the authors' code (Section 4); (2) an evaluation on two new datasets, the existing Emotion tweets and a dataset of 199 grocery-app reviews that I built (Sections 5 and 6); and (3) an LLM-as-judge check of whether EDA's edits keep their labels and stay readable (Section 6.3). Section 7 covers ethical considerations. The main finding is negative: I could not reproduce the gain on SST-2, and the picture on new data is mixed.

Original	whenever there is a problem it is resolved
Synonym replacement	whenever there is a problem it is decide
Random insertion	whenever there is trouble a problem it is resolved
Random swap	whenever there is it problem a is resolved
Random deletion	there is a problem is resolved

Table 1: One grocery-app review edited once by each EDA operation, using the authors' code ($\alpha = 0.1$). EDA first lower-cases the text and strips punctuation.

2 The Original Paper

2.1 What the paper does

Wei and Zou (2019)[1] start from a simple question: how can we generate extra sentences without changing their labels? For a training sentence they apply one of four random operations. *Synonym replacement* (SR) swaps n non-stopwords for WordNet synonyms (Miller, 1995). *Random insertion* (RI) inserts a synonym of a random word at a random position, n times. *Random swap* (RS) swaps two random words, n times. *Random deletion* (RD) drops each word with probability p. The number of edited words is $n = \alpha l$ for a sentence of l words (and $p = \alpha$), and n_{aug} augmented sentences are generated per original. Table 1 shows each operation applied to one review from my own data.

[1] https://aclanthology.org/D19-1670

2.2 How the paper evaluates EDA

The evaluation is straightforward. For each of five text classification datasets, the authors train a classifier with and without EDA and compare accuracy on the test set. They use two models, a CNN (Kim, 2014) and an LSTM-RNN, and random training subsets of 500, 2,000, 5,000 and all available sentences, because they expect EDA to help most when data is scarce. Every number is a mean over five random seeds. They also recommend settings for α and n_{aug} for each training size.

The paper's headline numbers are averages over the five datasets. For the CNN, accuracy rose from 78.6% to 80.7% at 500 sentences, from 85.6% to 86.4% at 2,000, from 87.7% to 88.3% at 5,000, and from 88.3% to 88.8% on the full sets. Averaged over both models, the gain was 3.0 points at 500 sentences and 0.8 on the full sets. The abstract's main claim is that, on average, training with EDA on only 50% of the data matched normal training on all of it (88.6% against 88.3%). Results for individual datasets, including SST-2, are shown only as curves, so these averages are the figures I can compare against numerically. The authors are candid about limits: they note that gains can be marginal when data is sufficient and may not appear with pre-trained models.

2.3 Why I chose this paper

Quality of the source. The paper appeared at EMNLP-IJCNLP 2019; EMNLP is ranked A* in the CORE ranking, as the unit's project page states (Macquarie University, 2026). Citation aggregators report roughly 1,800 to 2,000 citations: about 1,800 on Emergent Mind[2] and about 2,000 on Liner[3], both accessed on 9 November 2026. I have not checked Google Scholar, which the unit suggests and which may give a different count. Later work such as AEDA (Karimi et al., 2021) builds directly on EDA. **Feasibility.** The authors' code is public[4] and had about 1,700 GitHub stars and over 300 forks when I checked in 2026. I cloned it and ran its augmentation script on the bundled sample of 500 SST-2 sentences; it produced 2,500 sentences (the originals plus four edits each) within seconds. The datasets are public and small, and the method needs only a thesaurus, so I could run everything on free cloud hardware. **Interest.** The paper states its own limitations clearly, which makes its claims worth testing rather than just accepting.

[2] https://www.emergentmind.com/papers/1901.11196

[3] https://liner.com/review/eda-easy-data-augmentation-techniques-for-boosting-performance-on-text

[4] https://github.com/jasonwei20/eda_nlp

Dataset	Classes	Avg. words	Train	Test
SST-2	2	17	7,447	1,752
CR	2	18	4,082	452
SUBJ	2	21	9,000	1,000
TREC	6	9	5,452	500
PC	2	7	39,418	4,508

Table 2: The paper's five datasets (statistics from the paper's supplementary Table 5).

Label	Sentence
1	a stirring , funny and finally transporting re-imagining of beauty and the beast and 1930s horror fi...
0	apparently reassembled from the cutting-room floor of any given daytime soap .
0	they presume their audience wo n't sit still for a sociology lesson , however entertainingly present...

Table 3: The SST-2 file format: label (1 positive, 0 negative) and tokenised sentence.

3 The Original Datasets

The paper uses five benchmark datasets (Table 2), all short-text classification tasks: SST-2, movie-review sentence sentiment (Socher et al., 2013); CR, customer-review sentiment (Hu and Liu, 2004); SUBJ, subjective versus objective sentences (Pang and Lee, 2004); TREC, six question types (Li and Roth, 2002); and PC, pro versus con sentences (Ganapathibhotla and Liu, 2008). I replicated on SST-2 only.

Each dataset is a plain-text file with one example per line: a class label followed by the sentence. SST-2 is lower-cased and tokenised, with punctuation separated by spaces (Table 3). I used the standard sentence-level SST-2 files from a public compilation[5], which has 6,920 training, 872 development and 1,821 test sentences. This split differs slightly from the paper's own (7,447 train, 1,752 test).

[5] https://github.com/AcademiaSinicaNLPLab/sentiment_dataset

Train	α	n_{aug}	Normal	With EDA
500	0.05	16	500	8,500
2,000	0.05	8	2,000	18,000
5,000	0.1	4	5,000	25,000
6,920 (full)	0.1	4	6,920	34,600

Table 4: EDA settings used in the replication (the paper's recommendations) and the number of training sentences with and without EDA.

Train	Normal	+EDA	Δ	p	Wins
500	76.8 ± 1.0	76.3 ± 0.5	-0.5	0.14	1/5
2,000	80.0 ± 0.5	79.3 ± 0.7	-0.7	0.10	1/5
5,000	80.9 ± 0.7	80.4 ± 0.5	-0.5	0.12	1/5
Full (6,920)	81.2 ± 0.7	81.3 ± 0.8	$+0.1$	0.72	3/5

Table 5: SST-2 test accuracy (%), mean $\pm$ s.d. over five seeds, CNN. Δ is EDA minus normal; Wins is the number of seeds in which EDA beat normal training.

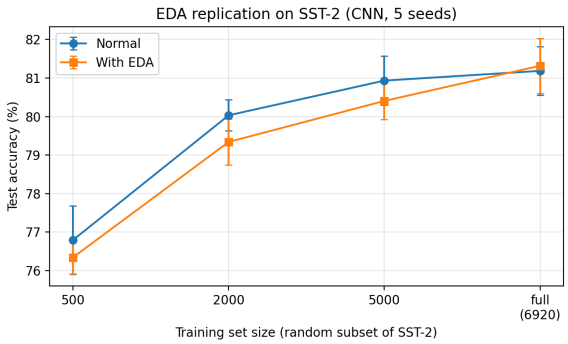


Figure 1: SST-2 accuracy with and without EDA (mean over 5 seeds, error bars $\pm$ 1 s.d.).

4 Replication of the Original Work

4.1 Setup and environment

All scripts are in my public repository[6]. I ran everything in Google Colab with Python 3 (the exact version was not recorded). For augmentation I used the authors' own `eda.py`, cloned at run time, with WordNet through NLTK. The authors' experiment scripts use Keras and the 840B GloVe file, which I did not want to depend on, so I re-implemented the CNN of the paper in PyTorch: static 300-dimensional GloVe vectors (Pennington et al., 2014), 128 filters of width 5, global max-pooling, a dense layer of 20 ReLU units and a softmax output, trained with Adam (learning rate 10^{-3}, batch size 64) and early stopping with patience 3 on the development set, restoring the best weights.

I followed the paper's protocol: random training subsets of 500, 2,000, 5,000 and all 6,920 sentences, five seeds, and the recommended EDA settings (α, n_{aug}) of (0.05, 16), (0.05, 8), (0.1, 4) and (0.1, 4). For each seed the normal and EDA runs share the same training subset, so the comparison is paired. With EDA, 500 sentences became 8,500 and the full set became 34,600. The whole experiment took about 50 minutes. I did not record the hardware type or the library versions in the Colab session that produced these results, which is a limitation: the repository's `requirements.txt` lists the packages but does not pin versions, so a re-run may differ slightly.

Table 4 lists the settings. I know of four differences from the original setup: PyTorch instead of Keras; the smaller GloVe 6B vectors (from `gensim-data`) instead of 840B; the standard SST-2 split; and a development set for early stopping.

[6] https://github.com/vinuthan25/COMP8240-EDA-reproduction

4.2 Results

Table 5 and Figure 1 show the results. EDA did not improve accuracy at any size. It was lower at 500, 2,000 and 5,000 sentences, by 0.5 to 0.7 points, and level on the full set. In each of those three sizes EDA beat normal training in only one of five seeds. None of the differences is significant (paired t-test, p between 0.10 and 0.72).

4.3 How close are the results, and why not closer?

My results do not match the paper. For the CNN, the paper reports average gains of +2.1, +0.8, +0.6 and +0.5 points at the four sizes; I measured -0.5, -0.7, -0.5 and $+0.1$. The two are not directly comparable, because the paper's numbers are averages over five datasets while mine come from SST-2 alone, whose per-dataset result the paper gives only as a curve. Still, the sign is wrong, and the biggest expected gain, at 500 sentences, is absent.

I looked for the cause but cannot name one with confidence. *Noise:* a test set of 1,821 sentences gives a standard error of roughly 0.9 points for a single run, and my seed-to-seed standard deviations were 0.5 to 1.0 points, so five seeds can only detect fairly large effects. *Setup differences:* the four differences listed above could matter, and my baseline CNN may be weaker than the authors'. The

paper's Figure 1 marks the best published CNN scores as a reference, and I did not tune hyperparameters. *Paper's own caveats:* the authors say gains shrink with enough data and may vanish with pre-trained models, and my CNN uses pre-trained (frozen) GloVe vectors, as theirs does. *Label noise:* the judge experiment in Section 6.3 shows that about 95% of EDA's edits keep their label, so noisy labels are unlikely to be the main problem. I did not run per-operation ablations or the paper's RNN, so I cannot rule out other explanations.

4.4 Code and repository

The repository has six folders of code and data, plus a `report/` folder holding this report. `replication/` holds the SST-2 script, its Colab notebook and the raw results; `data_collection/` holds the scraping, cleaning and sampling code for the reviews; `new_data/` holds the experiment code, the labelled review data and the results on both new datasets; `judge/` holds the 80 rated pairs, the rubric and the script that generated them; `analysis/` holds the script that recomputes the statistics in this report; and `docs/` holds the labelling guidelines. The code is released under the MIT licence; the review texts remain their authors' property. The README explains how to run each part: open the notebook in Colab and choose "Run all" (the new-data notebook asks for the review CSV). Package versions were not pinned, and because EDA is random and GPUs are not fully deterministic, a re-run will give slightly different numbers.

5 Construction of New Data

5.1 Existing dataset: Emotion

I chose Emotion (Saravia et al., 2018)[7] because it differs from the paper's data in the ways I care about: six classes instead of mostly two, noisy short tweets, and very uneven classes. It has 20,000 English tweets labelled with one of six emotions, split 16,000 / 2,000 / 2,000 into training, validation and test sets. In the training set joy has 5,362 tweets (33.5%), sadness 4,666 (29.2%), anger 2,159 (13.5%), fear 1,937 (12.1%), love 1,304 (8.2%) and surprise only 572 (3.6%). The tweets are already lower-case, so the only processing was EDA's own cleaning and the same pipeline as for SST-2 (Table 6 shows examples). I trained on random subsets of 500, 2,000 and 5,000 tweets; I left out the full 16,000 because EDA would have produced 64,000 sentences per run and I was short of time. I used the official validation set for early stopping and the official test set. My proposal also named IMDb, AG News, Yelp Polarity and Amazon Fine Food Reviews as candidate datasets, but time ran out before I could use them, so the evidence on generalisation rests on one existing dataset and one small new one.

[7] https://huggingface.co/datasets/dair-ai/emotion

5.2 New dataset: grocery-app reviews

Collection and processing. I wanted a domain not covered by the usual sentiment benchmarks: reviews of Australian supermarket shopping apps, which mix short, blunt complaints with longer, more detailed ones. I collected the newest 600 reviews from the Australian Google Play store for each of the Woolworths (`com.woolworths`) and Coles (`com.coles.android.shopmate`) apps with the `google-play-scraper` library (store country Australia, language English, newest first; the newest review is dated 8 October 2026), giving 1,200 reviews. Cleaning kept reviews of 4 to 150 words (1,048 left), removed duplicates (1,047) and dropped texts that were mostly non-English or emoji (1,044). I then drew 40 reviews per star rating, hid the stars, and removed one review that named a staff member, leaving 199. The labelled reviews span July 2025 to October 2026.

Annotation and agreement. Each review was labelled *negative*, *positive* or *mixed/neutral* from the text alone, with the star rating hidden. I did not label them from scratch. An AI assistant that could not see the star ratings suggested a first-pass label for every review, and I reviewed them all and changed five of the 199. This means the labels are *not* independent human annotation, and I had no second human annotator. As a measure of agreement, I compared the final labels with labels derived from the star ratings (1–2 stars negative, 4–5 positive, 3 stars excluded): they agree on 91.9% of the 149 reviews that qualify (Cohen's $\kappa = 0.84$). This is agreement with a noisy proxy, not inter-annotator agreement, and it should be read that way. One practical lesson: during my review I used a different code for the labels than the one I had been given (I treated 2 as "negative" and 0 as "neutral"), which made 81 labels look changed when only five were real disagreements. I only

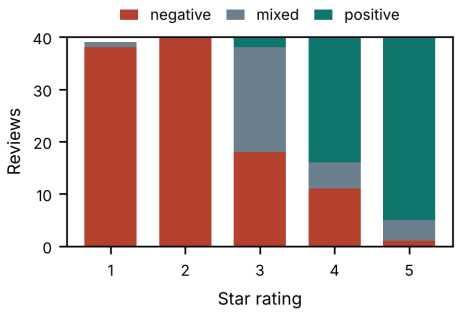


Figure 2: What the text says, by star rating, for the 199 grocery reviews.

Label	Text
Emotion (training set)	
sadness	i didnt feel humiliated
sadness	i can go from feeling so hopeless to so damned hopeful just from being around someone who cares and is aw...
anger	im grabbing a minute to post i feel greedy wrong
Grocery-app reviews	
negative	The "Masters" of disaster shoot themselves in foot once again. .. You have now changed your app and made ...
negative	Extremely difficult not able to finish my application
mixed	When this App is working to it's full potential, it's great. I particularly love the Watchlist feature. I...

Table 6: Examples from the two new datasets (long texts shortened).

caught this by checking the edits, which is a good argument for labelling with words instead of digits.

Properties and quality. The dataset has 108 negative, 61 positive and 30 mixed reviews (Woolworths 116, Coles 83). Reviews average 28.4 words (median 22, range 4 to 98), against 17 words per sentence in SST-2. A classifier that always predicts "negative" scores 54.3%. Star ratings are a noisy guide to sentiment here: of the 40 four-star reviews, 11 read as negative and 5 as mixed (Figure 2; counts in Appendix Table 11). Table 6 shows example texts from both new datasets. Compared with the original datasets, this one is much smaller (199 against thousands of examples), has a single annotator, and covers two apps. It has all the essential properties of the originals, namely real user text with sentiment labels, but it is too small for firm conclusions. I release 194 of the 199 reviews: five that disclose health circumstances are withheld (Section 7).

Train	Metric	Normal	+EDA	Δ	p
500	Accuracy	62.9	64.9	+2.1	<0.01
500	Macro-F1	42.2	45.1	+2.9	0.05
500	F1 (surprise)	2.1	5.1	+3.0	0.51
2,000	Accuracy	78.2	78.3	+0.1	0.84
2,000	Macro-F1	67.5	67.5	0.0	0.98
2,000	F1 (surprise)	36.5	36.9	+0.4	0.91
5,000	Accuracy	84.6	85.5	+0.9	0.06
5,000	Macro-F1	77.6	79.6	+2.0	0.03
5,000	F1 (surprise)	55.5	61.7	+6.2	0.05

Table 7: Emotion test results (%), mean over five seeds.

5.3 Experimental setup on the new data

I used the same CNN, EDA code and early-stopping rule as in Section 4, with a six-way (Emotion) or three-way (reviews) output layer. Emotion used five seeds and the paper's settings for each size. For the reviews I made ten stratified 80/20 train–test splits, holding out 15% of each training portion for early stopping (135 training reviews, 24 for development, 40 for testing) and used the settings for 500 sentences (0.05, 16). Besides accuracy I report macro-F1 and the F1 of the rarest class (surprise, mixed), because both datasets are imbalanced. All p-values are uncorrected paired t-tests.

6 Results on New Data

6.1 EDA on Emotion and on the grocery reviews

Table 7 and Figure 3 show Emotion. EDA improved accuracy at 500 tweets by 2.1 points ($p < 0.01$) and macro-F1 by 2.9 ($p = 0.05$). At 2,000 tweets there was no difference at all. At 5,000 tweets accuracy rose by 0.9 points ($p = 0.06$) while macro-F1 rose by 2.0 ($p = 0.03$), and the F1 of the rare class "surprise" by 6.2 ($p = 0.05$), which suggests that EDA mainly helped the smaller classes. At 500 tweets the models barely learned "surprise" at all (F1 of 2.1 and 5.1), and the difference is not significant.

On the grocery reviews (Table 8) EDA made no significant difference: accuracy rose from 69.8% to 70.5% ($p = 0.56$) and macro-F1 from 49.1 to 51.6 ($p = 0.16$). Accuracy is well above the 54.3% majority baseline, but without EDA the model never predicted the "mixed" class (F1 of 0). With EDA it found some mixed reviews in a few splits (F1 5.7), but with only about six mixed reviews in each test set this is not reliable.

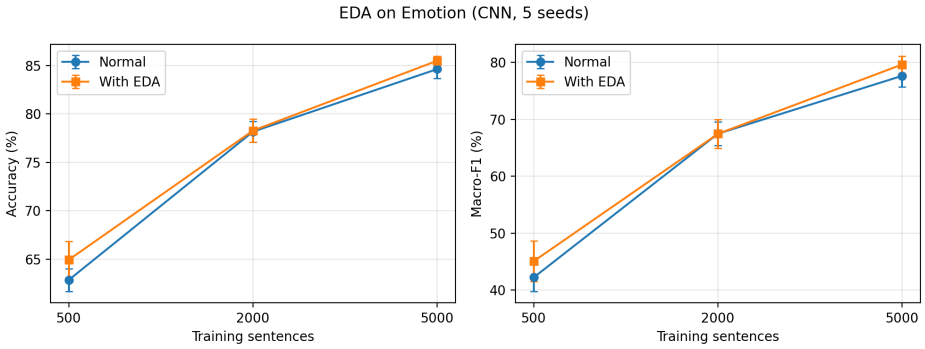


Figure 3: EDA on Emotion: accuracy and macro-F1 with and without EDA (mean over five seeds, error bars ± 1 s.d.).

Metric	Normal	+EDA	Δ	p
Accuracy	69.8	70.5	+0.8	0.56
Macro-F1	49.1	51.6	+2.4	0.16
F1 (mixed)	0.0	5.7	+5.7	0.17

Table 8: Grocery-review test results (%) on all 199 reviews, mean over ten stratified splits of 40 test reviews each.

6.2 How the results relate to the replication

Where the evaluation differs from the replication, the differences matter for comparing the numbers: (i) I added macro-F1 and rare-class F1 alongside accuracy; (ii) the datasets have six and three classes rather than two; (iii) Emotion used the official test split and sizes up to 5,000, while the reviews used ten repeated splits with tiny test sets; and (iv) the replication used five seeds on a fixed test set. With those caveats, the SST-2 replication and the new data disagree. On SST-2 EDA never helped, while on Emotion it helped at 500 tweets and, for macro-F1, at 5,000. The gain at 500 tweets (+2.1 accuracy) is the same size as the paper's average gain for the CNN at that size, so the paper's claim about small training sets is supported there, but the pattern is not monotonic, since 2,000 tweets showed nothing. On the reviews there is no sign of a benefit. With nine metric comparisons on Emotion and uncorrected tests, a few results near $p = 0.05$ could be chance, so I treat the Emotion gains as suggestive rather than established. Overall, the paper's claims did not carry over consistently to new data.

6.3 LLM-as-judge

How it fitted into the evaluation. Accuracy tells me whether a classifier trained with EDA does better, but not whether EDA's edits are *valid training examples*. The paper itself acknowledges that an edit could change a sentence's label, and checks this only visually, with a t-SNE plot on one dataset. I used an LLM as a judge to check it directly, following the idea of using strong language models as scalable evaluators of text (Zheng et al., 2023).

Operation	Label kept	Fluency	Human fl.
Synonym replacement	90%	3.10	3.0
Random insertion	100%	2.70	3.3
Random swap	95%	2.05	2.6
Random deletion	95%	3.35	4.0
All (80 pairs)	95%	2.80	3.23

Table 9: LLM-as-judge results, 20 pairs per operation. Label kept and fluency are the judge's ratings (1–5 scale). "Human fl." is my own mean fluency rating on 10 pairs per operation.

Method. I drew 10 random original texts from each dataset (6 to 40 words, excluding the reviews flagged for privacy) and edited each once with each of the four operations, using the authors' functions ($\alpha = 0.1$), giving 80 pairs. The judge, an LLM chat assistant used through its web interface in a single session on 9 November 2026 (the exact model version was not recorded), saw the dataset, the label, the cleaned original and the edit, and answered two questions for each pair: does the edit keep the same label (1/0), and how fluent is it (1 unreadable to 5 natural)? The judge did not know which operation made each edit. I rated a random 40 of the pairs myself first, without seeing the judge's ratings, so that I could measure agreement. The exact rubric is in the repository.

Results. The judge rated 76 of 80 edits (95%) as keeping their label, with a mean fluency of 2.8 (Table 9; examples in Table 10). The differences in label preservation between operations are one

Op.	Original (O) and edited (E)	Me	Judge
Synonym	*O:* why did u add soy to the bakery bread *E:* why did u add soya to the bakery bread	1	1
Synonym	*O:* i feel that way considering most people are pretending to be the way they are and very very few are being sincere *E:* i feel that fashion considering most people are guess to be the fashion they are and very very few are being sincere	0	0
Swap	*O:* why did u add soy to the bakery bread *E:* why did u bakery soy to the add bread	1	0
Synonym	*O:* fantastic to use really convenient and has never failed me *E:* fantastic to employment really convenient and has never failed me	0	1
Deletion	*O:* i get logged out every time i open the app *E:* i get out time i open the app	–	0

Table 10: Examples of judged pairs: label kept (1) or not (0), by me and by the judge (– means I did not rate that pair). The first row is a clean edit; the second is a case where both of us said the label was lost; the next two are the only disagreements among my 40 pairs; the last is a deletion that removed the key word.

or two pairs and should not be read as meaningful. Fluency does differ: random swap, which scrambles word order, was worst (2.05) and random deletion best (3.35). On the 40 pairs I also rated, the judge and I agreed on the label for 95% of pairs (38 of 40; we disagreed on two). Cohen's κ was only 0.47, but that is an artefact of both of us rating nearly everything "kept". On fluency, 85% of our ratings were within one point (Spearman 0.64), and the judge was stricter by about half a point (2.77 against my 3.23 on the same pairs). The ranking of operations was the same for both of us at the extremes.

What this adds. Edits mostly keep their labels, so label noise from EDA is unlikely to explain the weak results, although the sentences are often awkward. *Limitations:* the sample is small (80 pairs); it covers only joy, love and sadness, and negative and positive, with no rare classes; several originals do not clearly show their dataset label, so "kept" means "same signal as the original"; the judge was a single chat session and cannot be rerun exactly; and the same assistant suggested the first-pass review labels, so the judge is not fully independent there.

6.4 Threats to validity

These results come with caveats. *Statistical power:* five seeds on SST-2 and Emotion, ten small splits on the reviews and 80 judged pairs mean that only large effects can be detected, and the p-values are uncorrected. *Setup:* the replication differs from the original in framework, embeddings, data split and early stopping; I used one architecture (the CNN) without tuning it, and I did not run the paper's RNN or the other four datasets. *Labels:* the review labels were AI-suggested and reviewed by one person, so they are not independent human annotation, and agreement was measured against a proxy, the star ratings. *The judge:* it was a single session that cannot be repeated exactly, its sample covered only some classes, and the same assistant suggested the review labels. *Data:* the new dataset has 199 reviews from two apps, of which 194 are released. The conclusions should be read as indicative, not definitive.

7 Ethical Considerations

7.1 The original paper

The EDA paper has no ethics statement. This is not surprising for a 2019 paper about a simple technique, but several issues are worth naming. It does not say how the text in its datasets (customer reviews, movie reviews, pro/con sentences) was collected, or whether the people who wrote it could be identified or had consented. It acknowledges that an edit may change a label, and notes that deep learning is robust to label noise, yet it checks this only visually on one dataset; a mislabelled training example in a sentiment or emotion model is a quiet way for errors to enter. Positively, the paper reports its limitations plainly and releases code, which supports honest reporting and transparency. It does not discuss who could be affected if such a classifier were used on real users' text.

7.2 If I published research on the new data

Australian researchers, including students, are expected to follow the Australian Code for the Responsible Conduct of Research, whose eight principles are honesty, rigour, transparency, fairness, respect, recognition of Aboriginal and Torres Strait Islander peoples' rights, accountability and promotion of responsible practice (NHMRC, ARC and

Universities Australia, 2018). I apply them to my work below.

Honesty and transparency. I would disclose that the review labels were AI-suggested and reviewed by me (five of 199 changed) and that the agreement figure is against a proxy. I would state that the experiments used 199 reviews while the public file has 194, and report the failed replication and the non-significant results as they are, without selecting only the favourable comparisons.

Rigour. The samples are small, the p-values are uncorrected, the test sets for the reviews have about 40 examples, and the judge was one chat session. A published version would need a larger dataset, a second human annotator, a multiple-comparison correction and an automated, repeatable judge.

Respect and privacy. The reviews are public and I collected no usernames, but public is not the same as harmless: a review's text can be searched back to its author, and some reviewers disclosed health circumstances. I withheld five such reviews from the public release and removed one that named a staff member. The National Statement on Ethical Conduct in Human Research (NHMRC, ARC and Universities Australia, 2025) defines human research broadly (it includes research about people's data), and university guidance interpreting it says that research using only existing, publicly available and non-identifiable data of low risk may be exempt from review (Charles Darwin University, n.d.). My data is public but not guaranteed non-identifiable, so I would seek a determination from the university before publishing instead of assuming an exemption. I also used an unofficial scraping library and did not check Google Play's terms of use, which I would need to do first.

Fairness. The data covers two apps, English only, and a self-selected group of reviewers. Withholding the five health-related reviews protects those people but also removes the voices of some disabled users, who may be most affected by app accessibility problems. This is a real trade-off between privacy and representation, and a paper would need to say how it was handled.

Recognition, accountability and AI use. I credit the EDA authors, the dataset creators and the tool builders in the references. An AI assistant helped with code, analysis, drafting, label suggestions and the judge. Macquarie's policy on the responsible and ethical use of AI applies to students and expects integrity risks to be managed under the university's integrity policies (Macquarie University, n.d.), so I disclose this use and take responsibility for the content. The repository's README also records it.

Open access. Releasing code and data supports reproducibility, which the Code promotes, but it conflicts with privacy for user-generated text. My solution, releasing a reduced dataset and documenting what was withheld and why, keeps most of the benefit while limiting harm, but it means the released data cannot reproduce my exact numbers.

8 Conclusions and Reflections

I could only partly reproduce Wei and Zou (2019). On SST-2, with the authors' own augmentation code and the paper's CNN and protocol, EDA gave no gain: the paper's average improvement of +2.1 points at 500 sentences became -0.5 here, and none of my differences was significant. The paper's own caveats, that gains can be marginal and may not appear with pre-trained models, are consistent with this, but I could not isolate a cause. On new data the approach generalised only partly: on Emotion EDA helped at 500 tweets and, for macro-F1 and the rare class, at 5,000, but not at 2,000; on 199 grocery reviews it made no significant difference. The LLM-as-judge shows that EDA's edits keep their labels about 95% of the time but are often awkward, especially random swap.

What I would do differently: (1) Label the reviews with a second human annotator and with words rather than digits; a measured inter-annotator agreement would have been far stronger than agreement with star ratings. (2) Collect more reviews, so that the rare "mixed" class has more than a handful of test examples. (3) Draw the judge sample stratified by class, so that rare classes are covered. (4) Run per-operation ablations (my proposal promised a check on random deletion, which I only examined through the judge), the paper's RNN, other embeddings and a tuned baseline, to find out why SST-2 shows no gain. (5) Record library versions and hardware from the start, which I failed to do. (6) Report a multiple-comparison correction from the start. Annotation was harder than I expected, not because the reviews are ambiguous, but because small coding choices can silently change a whole dataset.

References

Charles Darwin University. n.d. Human research ethics approval procedures. University guidance.

Murthy Ganapathibhotla and Bing Liu. 2008. Mining opinions in comparative sentences. In *Proceedings of the 22nd International Conference on Computational Linguistics (Coling 2008)*, pages 241–248, Manchester, UK.

Minqing Hu and Bing Liu. 2004. Mining and summarizing customer reviews. In *Proceedings of the Tenth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 168–177. ACM.

Akbar Karimi, Leonardo Rossi, and Andrea Prati. 2021. AEDA: An easier data augmentation technique for text classification. In *Findings of the Association for Computational Linguistics: EMNLP 2021*, Punta Cana, Dominican Republic. Association for Computational Linguistics.

Yoon Kim. 2014. Convolutional neural networks for sentence classification. *arXiv preprint arXiv:1408.5882*.

Xin Li and Dan Roth. 2002. Learning question classifiers. In *COLING 2002: The 19th International Conference on Computational Linguistics*.

Macquarie University. 2026. Overall goal and assessment criteria. COMP8240 unit page, iLearn.

Macquarie University. n.d. Responsible and ethical use of artificial intelligence policy. University policy.

George A. Miller. 1995. WordNet: A lexical database for English. *Communications of the ACM*, 38(11):39–41.

NHMRC, ARC and Universities Australia. 2018. Australian code for the responsible conduct of research. Developed by the National Health and Medical Research Council, the Australian Research Council and Universities Australia.

NHMRC, ARC and Universities Australia. 2025. National statement on ethical conduct in human research.

Bo Pang and Lillian Lee. 2004. A sentimental education: Sentiment analysis using subjectivity summarization based on minimum cuts. In *Proceedings of the 42nd Annual Meeting of the Association for Computational Linguistics*, pages 271–278, Barcelona, Spain.

Jeffrey Pennington, Richard Socher, and Christopher D. Manning. 2014. GloVe: Global vectors for word representation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1532–1543.

Elvis Saravia, Hsien-Chi Toby Liu, Yen-Hao Huang, Junlin Wu, and Yi-Shin Chen. 2018. CARER: Contextualized affect representations for emotion recognition. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 3687–3697, Brussels, Belgium.

Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D. Manning, Andrew Y. Ng, and Christopher Potts. 2013. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing*, pages 1631–1642, Seattle, Washington, USA. Association for Computational Linguistics.

Jason Wei and Kai Zou. 2019. EDA: Easy data augmentation techniques for boosting performance on text classification tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 6382–6388, Hong Kong, China. Association for Computational Linguistics.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*.

A Supplementary Material

Stars	Negative	Mixed	Positive	Total
1	38	1	0	39
2	40	0	0	40
3	18	20	2	40
4	11	5	24	40
5	1	4	35	40

Table 11: Final labels by star rating for the 199 grocery reviews. Star ratings and text often disagree, especially at 3 and 4 stars.

Train	Run	Seed 0	Seed 1	Seed 2	Seed 3	Seed 4
500	Normal	75.5	77.6	76.2	76.8	77.9
	+EDA	75.8	76.8	75.9	76.3	76.8
2,000	Normal	79.8	80.3	79.9	79.5	80.6
	+EDA	78.3	79.3	80.1	79.4	79.6
5,000	Normal	81.2	81.3	81.7	80.6	79.9
	+EDA	80.9	80.3	80.9	79.6	80.3
Full	Normal	80.4	82.3	81.3	81.2	80.8
	+EDA	80.8	82.2	80.2	81.7	81.7

Table 12: SST-2 test accuracy (%) for every seed of the replication.

Labelling guideline for the reviews. Each review is labelled from its text alone, with the star rating hidden. *Positive:* the reviewer is mainly happy. *Negative:* the reviewer is mainly complaining, and a mild complaint still counts as negative. *Mixed/neutral:* praise and complaint together, or no real opinion. The AI assistant that suggested first-pass labels was given the same definitions and did not see the star ratings. The guideline is in `docs/` in the repository.

Judge rubric (summary). *Label preserved (1/0):* does the edited text still express the same emotion or sentiment as the original? It is 1 if the label-bearing words are still present and not contradicted, and 0 if the edit flips, removes or garbles them. *Fluency (1–5):* 1 unreadable, 2 hard to follow, 3 understandable but awkward, 4 one small oddity, 5 natural; missing punctuation and lower case are not penalised. The full rubric and all 80 rated pairs are in the repository folder `judge/`.